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Keypoint Detection for Identifying Body Joints using TensorFlow

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ABSTRACT

Keypoint Detection for the Body Joints is a process of finding and detecting all the important body joints of a human being. It takes an RGB image as input gives a list of keypoints as output. It can be applied for single-person body joints detection or multi-person body joints detection. The various architectures for detecting the human body joints can be compared using the Percentage of Detected Joints (PDJ), Object Keypoint Similarity (OKS) and mean Average Precision(mAP) evaluation metrics. The state-of-the-art body joint detection architectures like DeepPose, Higher Resolution Network and OpenPose architectures were studied and compared against each other. These architectures were evaluated on the COCO dataset for the 17 key body joints using the OKS and mAP evaluation metrics. The evaluation results on all the body joints were compared to each other for each architecture. The architectures were implemented using opensource implementations of the PyTorch and TensorFlow libraries. The HRNet architecture was the fastest and most accurate of all the architectures.

CCS CONCEPTS

• **Computing methodologies** → Artificial intelligence; Artificial intelligence; Computer vision; Computer vision problems; Interest point and salient region detections; Machine learning; Learning paradigms; Supervised learning; Supervised learning by regression.

KEYWORDS

Body Joints Detection, Keypoint Detection, Human Pose Detection

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1 INTRODUCTION

Keypoint is used to detect coordinates of an object in an image which can be of interest to a user. A good keypoint detection system is invariant to different transformation to the input image such as changes in brightness, skewness, occlusions, rotations and distortions. One of the most popular applications of key point detection is a Human Pose Estimation system which has tons of applications such as sports and fitness: to help athlete train better without the need of an actual physical trainer. It is also used for activity recognition which can be used to monitor crowds and easily maintain order in society. It can be used to gather lots of sports data from professional games without the use of expensive and uncomfortable sensors. It will bring a new way for human computer interaction. Human Pose Estimation will be the key area of our study for building a landmark detection system.

1.1 High Level Architecture of Landmark Detection

In a landmark detection problem, the machine learning model is required to output some coordinates. Each coordinate consists of a pair of values represented by (x, y) within the range of the pixel resolution of the image. Therefore, a landmark detection problem is basically a regression problem consisting of an RGB image as input, which is passed through a convolution neural network and 2k output numbers (where k is the number of key points). A pose detection system usually consists of 17 or more landmark points, whereas a face recognition system consists of at least 35 landmark points to gather as much information about the face possible.

1.2 Evaluation Metrics for Landmark Detection

An evaluation metric will let us compare two different pose detectors against the same test data to determine which one works better. It lets us infer the performance of the detector. There is not a single metric that can be used for all the machine learning problems; different tasks require different evaluation metrics. For example: a classification problem uses accuracy and confusion matrix as an evaluation metric while a regression problem uses mean absolute error and mean squared error as different evaluation metrics. The different evaluation metrics for a landmark detection and pose detection system are: Percentage of Detected Joints (PDJ) is an evaluation metric that considers a joint to be detected if the distance between the detected key point or joint and the true joint is less than a given threshold.

The threshold is usually 0.05*diagonal length of the person detected. This method sums up all the detected joints and neglects all the undetected joints to calculate the percentage of joints detected [1].

$$PDJ = \frac{\sum_{i=1}^{n} bool\ (d_i < 0.05 * diagonal)}{n} \quad (1)$$

In equation 1), di is the distance between the true joint and the predicted joint. Diagonal is the length of the diagonal. Bool is the true or false condition; and n is the number of key-points. OKS is an evaluation metric that is very similar to PDJ. It calculates the distance between the predicted and true joint but also considers a constant for each key point because the threshold for each key point needs to be different; for eg: the threshold for eyes needs to be less than the threshold for shoulders. It also considers the scale of the person which is the square root of the area under the bounding box.

$$OKS = exp\left(-\frac{d_i^2}{2s^2k_i^2}\right) \quad (2)$$

In equation 2), 'di' is the distance. 'S' is the scale and 'k' is the constant for each individual key point [12]. This metric was used by the COCO dataset. Different values for k_i are chosen according to the body joint. The value of OKS is between 0 and 1. If the value of this metric is 1 then the position of the predicted and true joints is exactly the same. This metric is used by the Average Precision metric which considers a key point to be detected if its OKS value is more than a given threshold. The threshold value is usually set more than 0.5 but less than 0.75 [2].

1.3 DeepPose

DeepPose is a pose estimation architecture proposed by researchers of Google in the year 2014. It uses a convolution neural network for estimating the poses of humans. It considers pose estimation as a deep neural network-based regression problem for the key body joints. It uses a cascade of such regressors to give highly precise key points. The different components of the DeepPose architecture consists of many components [9]. The Pose Vector is the output label of the deep neural network. It is represented using the variable y as shown in equation 3). All the k key points or joints are merged into a single vector of k elements.

$$\mathrm{y} = \left(\ldots, \mathrm{y}_\mathrm{i}^\mathrm{T}, \ldots\right)^\mathrm{T}, \mathrm{i}\ \varepsilon\ (1, \ldots, \mathrm{k}) \quad (3)$$

Each ith element of this vector packs two numbers within itself representing the (x, y) coordinates of the key point on the image. Each label of the training set consists of (x, y) where x is the input image and y are the output pose vector. The input image is cropped and resized according to the detected bounding box around the person. Since the image is resized the output pose vector labels need to be transformed as well otherwise the labels will be misinterpreted. The pose vector coordinates are normalized using the bounding box of the person [4]. The bounding box is represented using b = (b_c, b_w, b_h), where b_c is the center coordinates of the box and the other two are the width and height of the box. The output y label is normalized using the formula shown in equation 4).

$$N\ (y_i, b) = \begin{pmatrix} 1/b_w & 0 \\ 0 & 1/b_h \end{pmatrix} (y_i - b_c) \quad (4)$$

This gives us the normalized pose vector. The normalized input image is the cropped version of the image x done using the bounding box. The DeepPose architecture uses a CNN architecture to output the pose vector. It uses the AlexNet architecture as the CNN architecture as it was the best architecture for localization tasks at the time [13]. The first layer of the architecture is the input layer that inputs a 220x220 sized RGB image. The architecture then consists of 7 more layers consisting of convolutional, pooling and normalization layers. The convolutional layers use a stride of 4. The last few layers of the network are fully connected layers resulting in 2k points in reference to the k joints. The architecture is to be trained on 40 million parameters and it uses L2 loss function which basically performs mean squared error across all body joints. The input size of the image is fixed at 220x220 so it is difficult to get more accurate pose detections without the increase in resolution of the input. Increasing the size of the input will also increase the number of weights to be trained subsequently. Therefore, a cascade of deep neural networks-based pose regressors are used. This method uses multiple stages to get the final output pose vector. This method makes the accuracy of this architecture a lot better [3].

1.4 High Resolution Network

High Resolution Network popularly known as HRNet is one of the most popular architectures for Pose Estimation. This architecture was presented in the year 2019 at the famous computer vision CVPR conference. The main idea behind the HRNet was to maintain the resolution of the input throughout the network [10]. The researchers for the HRNet found out that a simple classification ConvNet cannot be used for pose estimation and segmentation tasks as well because the resolution of the input keeps on decreasing as we go deeper into the network and information about the positions of the key features are lost. Therefore, getting a high-resolution output is necessary for tasks other than classification like pose estimation and image segmentation. The paper stated that the first idea was to replace the last smaller layers of the convolution network are replaced with a dilated convolution layer but the major disadvantage of this approach was very computationally expensive. The paper stated that the second idea in the research community was to use a network that uses up sampling like the Hourglass Network [14]. This network consisted of a sequence convolutional layer connected to each other in series. A major disadvantage of this approach was that the spatial information is not very well represented in the output because up sampling part of the network is not able to recover the information about the positions of the features really well. The HRNet proposes to use a high-resolution network as its own backbone instead of using a classifier network which resulted in a decrease in resolution of the input. The HRNet aims to keep the resolution high throughout the whole network instead of first decreasing the resolution and then increasing it [4].

The earlier classifier networks were used to connect the high-resolution conv layers to the low-resolution conv layers in series [5]. The HRNet connects these different levels of convolutions in parallel. The high-resolution convolutions consist of more spatial information while the low-resolution convolutions have a higher effective receptive field. The HRNet uses 'repeated-fusions' to connect the information from the low resolution to the high resolution. Repeated fusions are the parallel connection between various levels of convolutions. The high-resolution convolution is connected to another high-resolution convolution directly with an identity operation. The medium and low-resolution convolutions are connected to the high-resolution convolutions with the 'bilinear up sampling' operation to increase the resolution and is then passed through a 1x1 conv layer to align the number of channels. Bilinear up sampling is preferred to convolutions for increasing the resolution because it is computationally less expensive. The high resolution to the medium and medium to the low resolution is connected using a 3x3 conv layer with a stride of 2. The low-to-low resolution is connected to each other using identity connection. The HRNet maintains high resolution throughout the network by connecting the multi-resolution layers in parallel. The architecture consists of 4 blocks and the last stage of each block consists of repeated-fusion connections. The depth of each resolution is a multiple of 'c' and the value of c is usually 32 or 48 depending on the application. The networks complexity and inference time is less than ResNet. Human Pose Estimation only uses the output of the high-resolution to produce heat maps for the body joints [6].

1.5 OpenPose

Open Pose is a state-of-the-art multi-person human pose estimation architecture. It was proposed by researchers at Carnegie Mellon University (CMU) in the year 2018 [11]. It uses a bottom-up approach for human pose estimation meaning that it first predicts all the body joints of all the people in the input image and then groups the detected key points together. It is a more complex method than top-down which first detects a person and then predicts the body joints of each person individually. It uses the first ten convolution layers from the VGG architecture as the feature extractor [7]. This backbone network takes the RGB image as input and gives a feature map as output represented by F. The Open Pose architecture consists of two branches with multiple stages stacked in front of one other. The stacking of stages increases the accuracy of the output. The architecture produces the 'Confidence Maps' and the 'Parts Affinity Fields' as the output in its two branches. The confidence maps are heatmaps for the different body joints like for the elbows, shoulders, knees etc. [8]. The second branch of the network is for outputting the 'Parts Affinity Fields' which detects the vectors of the paired joints. The confidence maps produced by the first branch are represented by 'S' and the parts affinity fields are represented by 'L'. The consecutive stages take the outputs from the previous stage, that is confidence map (S) and parts affinity field (L), along with the feature map F as inputs and joins them together to pass through its two branches until it reaches the last stage. The Open Pose architecture consists of 6 stages. The earlier stages are not very perfect in its prediction of the body joints but as we move along the network to the later stages it starts to make more confident and more accurate predictions and doesn't make such misclassifications. After the final stage the confidence maps and the parts affinity field are used to output the key body points using the process of greedy inference.

2 LITERATURE SURVEY

DeepPose is a pose estimation architecture proposed by researchers of Google in the year 2014. It uses a convolution neural network for estimating the poses of humans. It considers pose estimation as a deep neural network-based regression problem for the key body joints. It uses a cascade of such regressors to give highly precise key points. This method uses multiple stages to get the final output pose vector. This method makes the accuracy of this architecture a lot better [9]. High Resolution Network was presented in the year 2019 at the famous computer vision CVPR conference. The HRNet aims to keep the resolution high throughout the whole network instead of first decreasing the resolution and then increasing it. The HRNet uses 'repeated-fusions' to connect the information from the low resolution to the high resolution. The networks complexity and inference time is less than ResNet. Human Pose Estimation only uses the output of the high-resolution to produce heat maps for the body joints [10]. Open Pose was proposed by researchers at Carnegie Mellon University (CMU) in the year 2018. It uses a bottom-up approach for human pose estimation. It uses the first ten convolution layers from the VGG architecture as the feature extractor. It consists of two branches with multiple stages stacked in front of one other. The stacking of stages increases the accuracy of the output. The earlier stages are not very perfect in its prediction of the body joints but the later stages start to make more accurate predictions and don't make such misclassifications. After the final stage the confidence maps and the parts affinity field are used to output the key body points using the process of greedy inference [11]. Table 1 shows all the architectures used and the performance of each architecture on the MPII dataset.

3 DESCRIPTION OF DATASET AND EXPERIMENTATION SETUP

We have implemented all the state-of-the-art architectures for human pose estimation on two-dimensional images using deep neural network to compare the results of these architectures against each other. The percentage of detected joints and mean average precision using object key point similarity were used to compare the performances of the models. Each model was run on two datasets and the results for all joints were obtained. The results were analyzed to infer which key point on the body is easier to detect and which one is difficult for all the models. The average of the results of all the joints was used to compare every model against each other. The model giving the best performance was used for inference.

3.1 Methodology and Dataset Description

The DeepPose and HRNet architectures were implemented using mmPose which is an open-source pose detection tool built on top of PyTorch. The mmPose tool was built by the open mmLab which is an open-source computer vision organization. The COCO dataset was used for evaluating the performance of the models. COCO stands for common objects in context which is a large dataset that provides with vision-based dataset for image recognition, object detection, object segmentation and various key point detection-based applications like facial detection or pose detection. The COCO dataset contains a diverse set of images which lets us get a generalized performance of our architectures. The key point COCO dataset consists of 17 key points indicating the 17 body joints like head, both shoulders, both knees, ankles etc. Each key point is represented by (x, y, v); where x and y represent the coordinates and v represents the visibility of the joint i.e., whether it is visible or

Table 1: Different Architectures Used

S.No.	Author	Architecture Used	Result (average precision AP) on MPII	Ref.
1.	A. Toshev and C. Szegedy	Deep Pose	85.0	[9]
2.	J. Wang et al	Higher Resolution Network	92.5	[10]
3.	Z. Cao, G. Hidalgo, T. Simon	OpenPose	61.8	[11]

not. Though COCO, contains images of a lot of categories, but it contains the key point annotations of human beings. The COCO key point is a very large dataset so a subset of the dataset is used for inference.

3.2 Implementation and Testing

The implementation of the key point detection of the joints of the body for human pose estimation was done on Google Colab. The COCO dataset is a huge dataset and the format of the dataset is complicated and contains a lot of unnecessary information. So, the authors clean the dataset and convert it to a format which is suitable for inference and evaluation of the model. The authors installed the CocoDataset library which lets us download the COCO Dataset. The authors download the annotations of the 2017 version of the dataset which is a JSON file. Two thousand images are downloaded by importing the coco_dataset library. The annotations are converted from a JSON file to a list of annotations each containing the segmentation and keypoints for each person. The annotations for each individual image id are extracted and are converted to a list of keypoints. These keypoints combined with the bounding box annotations and the image id are structured into a list of python dictionaries. The list of dictionaries is then converted into a panda's data frame to create a structured format of the dataset that can be easily accessed for evaluation against the prediction set. The joints which are not labelled are signified by -1. The Object Key-point Similarity and the mean Average Precision metrics are implemented using Pandas and NumPy from scratch. The Object Key Point Similarity calculates the Euclidean distance between two body joints and divides it by its area under segmentation of the area. Instead of using the diagonal length: a different key point index is used for each body joint as some joints are easier to detect than others. The DeepPose architecture is implemented using the mmPose library. DeepPose, like any other top-down pose detection architecture, needs a person detection object using which it will first detect the person in the input image and then detect the key points. The DeepPose implementation is available for both: COCO and MPII dataset. This implementation inputs an image of 256x192 resolution. The predictor data frame is constructed for the COCO dataset by performing inference on every image and storing the result in a list of dictionaries. The inference on all of the COCO dataset takes a little more than 1 minute. The prediction data frame is constructed using the list. The predictions are evaluated against the original ground truth labels using the custom-built evaluation functions. The HR Network is implemented using the mmPose library. MmPose only has the implementation of the MPII dataset for the architecture. The HRNet architecture inputs an image of size 256x256. The inference on the complete dataset takes 10 minutes.

Figure 1: Inference using the OpenPose architecture

The evaluations on the resulted predictions are done using the evaluation function. The OpenPose architecture is implemented using an opensource implementation of OpenPose built using TensorFlow. A pose detector function is created to process the inference from the OpenPose architecture. The results of inference are added to a list of dictionaries. The list is converted to a data frame of predictions. The inference for the COCO dataset takes approximately 2 minutes. The evaluation is done using the evaluation function. The evaluation is performed on the detections on the MPII dataset. Inference is performed to test a model on a single image to visually demonstrate how the results of the metrics reflect in real world detections. The skeleton-like structure for every pair of body joint is constructed using OpenCV. The OpenPose architecture is used for detecting body joints.

Inference is performed to test a model on a single image to visually demonstrate how the results of the metrics reflect in real-world detections as shown in Figure 1. The skeleton-like structure for every pair of body joint is constructed using OpenCV. The OpenPose architecture is used for detecting the body joints. The architecture detects all the joints accurately and connects them together to form a skeleton-like structure.

4 RESULTS AND DISCUSSION

The DeepPose and OpenPose architectures prove to be a little faster than the HRNet architecture. The architectures are compared to each other based on the PDJ, OKS and mAP metrics. The average of all the results obtained for the MPII Dataset is shown in the Figure 2, Figure 3 and Figure 4.

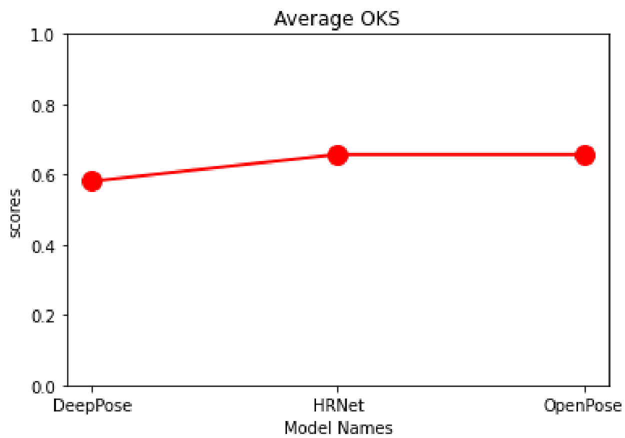

Figure 2: Plot for average OKS for the DeepPose, HRNet and OpenPose architectures with the plots shown from left to right.

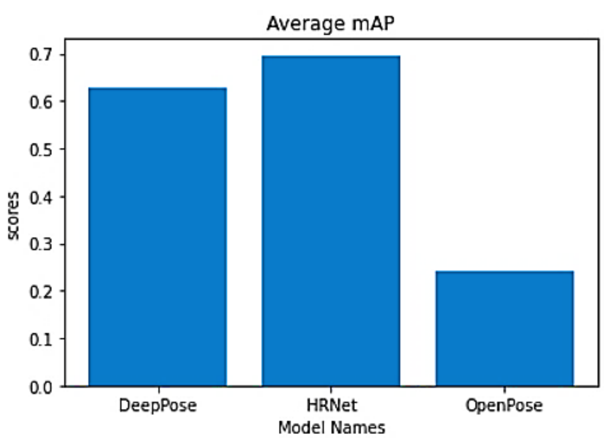

Figure 3: Bar Graph for Average mAP evaluations on all the architectures using the MPII dataset.

The average for a given metric is calculated by taking the mean of the results of all the joints over all the samples in the complete dataset. Thus, the average of a metric is a single number depicting the general performance of an architecture on all the joints. The inference on the complete dataset using the DeepPose architecture takes 6 minutes. The inference using the High-Resolution Network takes 10 minutes on the complete COCO dataset. The evaluation using the OpenPose architecture takes 5 minutes on the dataset.

The Figure 2, Figure 3 and Figure 4 shows that the HRNet gives the most optimal accuracy on all the evaluation metrics and is consistent in its performance. The evaluation of the predictions of each individual joint is shown in Figure 5 and Figure 6.

The architecture performs well on big joints like shoulders but struggles a little on small joints like ankles. The CPM gives the next best performance and the OpenPose architecture gives the worst performance as expected.

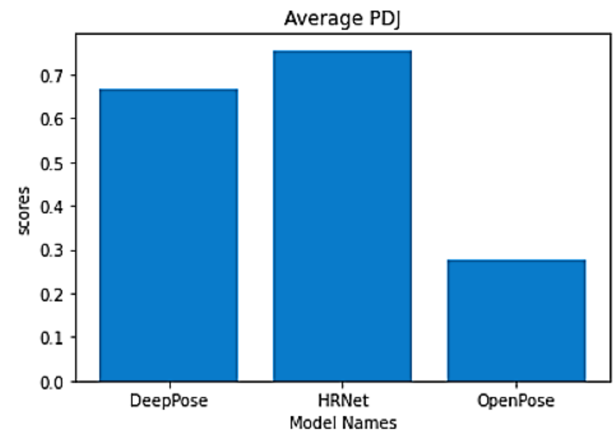

Figure 4: Bar Graph for average Percentage of Detected Joints on all the architectures.

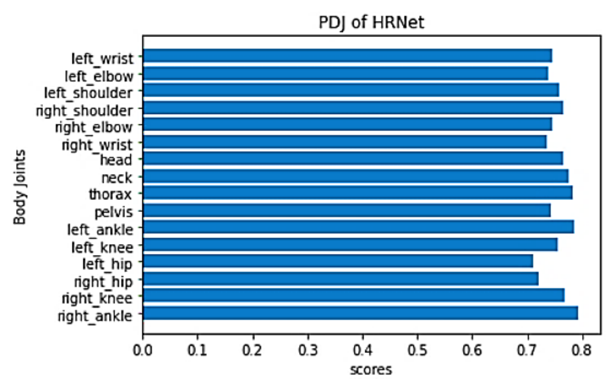

Figure 5: Horizontal Bar Graph for the Percentage of Detected Joints for the individual joints using the HRNet architecture.

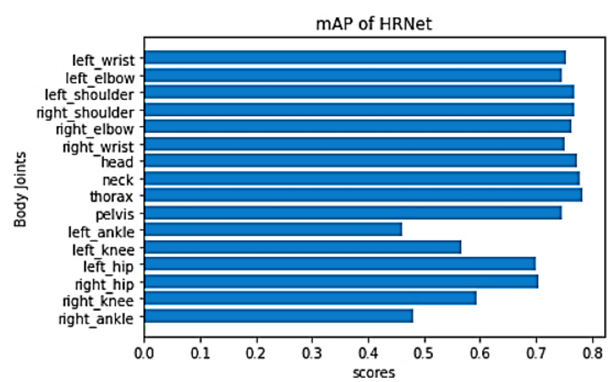

Figure 6: Plot for the evaluation of the HRNet architecture using the mAP evaluation metric on the individual joints

The mAP for all the architectures for each joint is shown in Table 2. HRNet performs better than rest of the architectures for all the joints. Figure 5 and Figure 6 shows that if the performance is measured irrespective of the size of a joint then the HRNet consistently

Table 2: The mAP on all the important joints using all the architectures on the MPII Dataset.

Joint	DeepPose	HRNet	OpenPose
RAnkle	0.3756	0.4798	0.1396
RKnee	0.5010	0.5931	0.1878
RHip	0.6158	0.7051	0.2104
LHip	0.5931	0.7002	0.1863
LKnee	0.4351	0.5676	0.1389
LAnkle	0.3288	0.4606	0.1013
Pelvis	0.7186	0.7462	0.2423
Thorax	0.7795	0.7838	0.4925
Neck	0.7668	0.7781	0.4280
Head	0.7413	0.7746	0.0354
RWrist	0.6676	0.7519	0.2126
RElbow	0.7150	0.7625	0.2934
Rshoulder	0.7406	0.7689	0.3798
Lshoulder	0.7420	0.7689	0.3366
LElbow	0.6555	0.7462	0.2530
Lwrist	0.693	0.7540	0.2317

performs well on all joints using the PDJ evaluation metric but if the size of a joint is considered then the performance drops a little for small joints as proved by the mAP metric.

5 CONCLUSION

A Human Pose Estimation system which has tons of applications such as sports and fitness: to help athlete train better without the need of an actual physical trainer. It is also used for activity recognition which can be used to monitor crowds and easily maintain order in the society. It can be used to gather lots of sports data from professional games without the use of expensive and uncomfortable sensors. It will bring a new way for human-computer interaction. Human Pose Estimation will be the key area of our study for building a landmark detection system. The human pose estimation system is used in sports analytics for analysis of sports data. The Human Pose Estimation can be used with the concept of Pose Tracking to increase the number of applications and use of this system which will help automate a lot of tasks like automated monitoring of security cameras of shops for identifying theft.

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